

Midterm Topics 24-25

Below is a list of topics to help you study for the Midterm Exam. The Midterm is over Units 1-3.

- Definition and justification of a function
- Domain, range, and properties of functions
- Composition of functions, taking the domain into account
- Inverse functions and the relationship between a function and its inverse
- Represent rate of change graphically to/from a scenario
- Function patterns
- Transformations of functions and graphs
- Quadratics: graphing by hand, identifying the y-intercept, vertex, roots, complex roots and inputs, complex inputs
- Definition of each conic section
- All four conic sections: converting general form to standard form and vice versa, graphing, and identifying attributes
- Modeling conic sections with matrices
- Simplifying expressions and solving equations with exponential functions
- Simplifying expressions and solving equations with logarithmic functions
- Creating a model for exponential growth or decay and solving for an unknown, including compound interest
- Graphs and equations of exponential and logarithmic functions
- Exponentials and logarithms as inverses, algebraically and graphically

Additional Information:

- Anything not listed above will NOT be tested on the Midterm.
- A calculator will NOT be allowed.
- Notes or other outside resources will NOT be allowed.
- You will have 45 minutes to complete the Midterm.
- The Midterm will be only one part; there are no open-response questions.